Table 1.1 List of some metals and their common ores with their chemical formula

Metal	Ore	Composition	Metal	Ore	Composition
Aluminum	Bauxite	$Al_2O_3.nH_2O$	Zinc	Zinc blende or Sphalerite	ZnS
	Diaspore	$Al_2O_3.H_2O$		Calamine	$ZnCO_3$
	Kaolinite	$Al_2Si_2O_5(OH)_4$		Zincite	ZnO
Iron	Haematite	Fe_2O_3	Lead	Galena	PbS
	Magnetite	Fe_3O_4		Anglesite	$PbSO_4$
	Siderite	$FeCO_3$		Cerrusite	$PbCO_3$
	Iron pyrite	FeS_2	Tin	Cassiterite (Tin stone)	SnO_2
	Limonite	$Fe_2O_3.3H_2O$	Silver	Silver glance (Argentite)	Ag_2S
Copper	Copper pyrite	$CuFeS_2$		Pyrargyrite (Ruby silver)	Ag_3SbS_3
	Copper glance	Cu_2S		Chlorargyrite (Horn Silver)	AgCl
	Cuprite	Cu_2O		Stefinite	Ag_5SbS_4
	Malachite	$CuCO_3.Cu(OH)_2$		Proustite	Ag_3AsS_3
	Azurite	$2CuCO_3.Cu(OH)_2$			

1.2 Concentration of ores

Generally, the ores are associated with nonmetallic impurities, rocky materials and siliceous matter which are collectively known as gangue. The preliminary step in metallurgical process is removal of these impurities. This removal process is known as concentration of ore. It increases the concentration of the metal of interest or its compound in the ore. Several methods are available for this process and the choice of method will depend on the nature of the ore, type of impurity and environmental factors. Some of the common methods of ore concentration are discussed below.

1.2.1 Gravity separation or Hydraulic wash

In this method, the ore having high specific gravity is separated from the gangue that has low specific gravity by simply washing with running water. Ore is crushed to a finely powdered form and treated with rapidly flowing current of water. During this process the lighter gangue particles are washed away by the running water. This method is generally applied to concentrate the native ore such as gold and oxide ores such as haematite (Fe_2O_3), tin stone (SnO_2) etc.

XII U1 Metallurgy - Jerald.indd 3 16-12-2022 18:52:57